

EXACT HISTORY DYNAMICS, DOMAIN-SAFE MEMORY, AND UNIVERSAL EXCESS PATH PRESSURE

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ABSTRACT. Conditional future laws are defined on the two-sided natural extension given the genuine past sigma-field. This yields an all-real-time Feller history semigroup and a uniform quenched rough invariance principle. The memory realization keeps the complete unresolved Hilbert space, adds only finite Riesz modes, and uses the causal Volterra sign.

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1. ROUND-EIGHT RECONSTRUCTION: PAST FILTRATION, EXACT UNRESOLVED DILATION, AND CAUSAL MEMORY

A quotient by stable leaves retains the future-determining coordinate and is therefore unsuitable for a nondegenerate conditional future. We work on the two-sided natural extension and condition on the genuine past sigma-field.

1.1. The natural extension and the past kernel. Let $(\Omega, \sigma, \nu_\Psi)$ be the two-sided Gibbs extension of the inducing system. Put

$$\mathcal{F}_0^- = \sigma(\omega_j : j \leq 0), \quad h = (\dots, \omega_{-1}, \omega_0).$$

The normalized transfer eigenfunction gives a g -kernel

$$g_\Psi(h, i) = \nu_\Psi(\omega_1 = i \mid \mathcal{F}_0^-(h)).$$

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It is a forward conditional law because the orientation is fixed by the natural extension.

Lemma 1.1 (Nondegenerate past disintegration). *If at least two admissible successors have positive $g_\Psi(h, \cdot)$, the conditional future given h is non-Dirac. The kernel depends continuously on h in the weighted past metric and satisfies the tower identity.*

Proof. The finite-cylinder conditional probabilities are ratios of positive transfer weights. The spectral variation estimate makes these ratios Cauchy as the known past grows, uniformly in the weighted metric. Their limit is g_Ψ . Multiplying successive kernels gives the regular conditional law of future cylinders, and summing over the first unknown symbol proves the tower identity. Positivity of two successors makes the first-step law non-Dirac. $\square$

Lift this kernel to the suspension by retaining the current roof age. Let $\mathcal{H}^-$ be the weighted space of pasts and ages.

Theorem 1.2 (All-real-time Feller history semigroup). *The renewal formula based on g_Ψ defines a conservative Feller semigroup $\mathcal{P}_t$ on $C_b(\mathcal{H}^-)$. It is constructed before any Ray compactification and satisfies Chapman–Kolmogorov for every real $t \geq 0$.*

Proof. Split a trajectory according to the number of completed roofs before time t . Finite horizon and the exponential roof moment make the renewal sum normally convergent. Continuity of each finite term follows from Lemma 1.1; the tail is uniform. Splitting at time s and using the tower identity gives $\mathcal{P}_{t+s} = \mathcal{P}_t \mathcal{P}_s$. Conservation follows by applying the formula to one. $\square$

Corollary 1.3 (Ray realization without circularity). *The resolvent of $\mathcal{P}_t$ has a Ray compactification carrying a right-continuous strong Markov realization.*

Proof. The already constructed conservative Feller resolvent separates the weighted past states and is normal under monotone convergence. The Ray construction therefore applies to this resolvent; no path regularity is used to define it. $\square$

1.2. Conditional enhanced homogenization. Let A be a centered additive observable and write the transfer Poisson solution χ_h on each past fiber. The martingale increments are

$$D_{n+1} = A \circ \sigma^n + \chi \circ \sigma^{n+1} - \chi \circ \sigma^n.$$

Theorem 1.4 (Uniform quenched rough invariance principle). *On every compact weighted past set outside a set of conditional probability $O(e^{-cL})$, the conditional laws of the rescaled path and its iterated integral converge uniformly to a Brownian rough path. The antisymmetric second level contains the deterministic area correction*

$$\mathfrak{a} = \frac{1}{2} \sum_{n \geq 1} \nu_\Psi(A \otimes A \circ \sigma^n - A \circ \sigma^n \otimes A).$$

Proof. The past-kernel transfer operators have the same spectral gap on every compact past chart, so the Poisson solutions and their conditional $L^{2+\delta}$ bounds are uniform. The martingale triangular-array invariance principle then holds uniformly. Burkholder estimates give tightness of the first and second levels. The coboundary terms vanish after scaling, while the predictable antisymmetric correlation sum converges absolutely to $\mathfrak{a}$. Exponential roof and distortion tails remove the exceptional histories. $\square$

1.3. The exact unresolved-space dilation. Let L be the closed microscopic generator on a Hilbert space $\mathcal{H} = V \oplus Q\mathcal{H}$, and let P be the orthogonal projection onto V . Its compressed resolvent is

$$C(z) = P(z - L)^{-1}P, \quad C(z)^{-1} = z - PLP - M(z),$$

where

$$M(z) = PLQ(z - QLQ)^{-1}QLP$$

is the exact unresolved transfer function. Let $Q\mathcal{H} = \mathcal{H}_{\text{cyc}} \oplus \mathcal{H}_\perp$, where $\mathcal{H}_{\text{cyc}}$ is the cyclic subspace seen by QLP and PLQ .

A finite Riesz–Potapov extraction of the selected transmission zeros gives an exact transfer decomposition

$$M(z) = M_{\text{res}}(z) + C_Z(z - A_Z)^{-1}B_Z$$

and a unitary realization map

$$U : \mathcal{H}_{\text{cyc}} \longrightarrow \mathcal{H}_{\text{res}} \oplus Z, \quad \dim Z < \infty.$$

The residual realization is chosen minimal; hence it has no selected transmission zero. Define

$$\tilde{\mathcal{H}} = V \oplus \mathcal{H}_\perp \oplus \mathcal{H}_{\text{res}} \oplus Z$$

and let $\tilde{L}$ be the unitary conjugate of L under $I_V \oplus I_{\mathcal{H}_\perp} \oplus U$.

Theorem 1.5 (Exact Riesz–Schur extraction with full unresolved dynamics). *The operator $\tilde{L}$ is fixed and closed, and*

$$P(z - \tilde{L})^{-1}P = C(z).$$

Its V -Schur complement is

$$z - PLP - M_{\text{res}}(z) - C_Z(z - A_Z)^{-1}B_Z.$$

Thus the finite state Z carries exactly the selected transmission-zero factor, while $\mathcal{H}_\perp \oplus \mathcal{H}_{\text{res}}$ retains every unresolved microscopic mode. No finite rational term is added on top of the old Q -resolvent. If the compressed resolvent is passive, the extraction can be chosen as a conservative Hilbert-space colligation.

Proof. On the cyclic subspace, the positive-real transfer M has a minimal Hilbert realization unique up to unitary equivalence. Factor the finite Blaschke–Potapov product generated by the selected Riesz zero projections from that realization. Each finite factor has a conservative state-space realization on Z , and the remaining outer transfer has a minimal realization

on $\mathcal{H}_{\text{res}}$. The cascade realization gives the displayed exact sum for M and is unitarily equivalent to the original cyclic realization. Extend this unitary by the identity on the invisible complement $\mathcal{H}_{\perp}$ and on V . Resolvents are invariant under unitary conjugacy, so the compressed resolvent is exactly $C(z)$. Applying the block inverse formula after the decomposition gives the stated Schur complement. Conservative factor and outer realizations preserve passivity. $\square$

1.4. The causal Volterra sign. For a resolved correlation satisfying

$$C'(t) = AC(t) + (K * C)(t), \quad C(0) = I,$$

Laplace transformation gives the causal memory transform

$$\widehat{K}(z) = zI - A - \widehat{C}(z)^{-1}.$$

Theorem 1.6 (Residual-memory decay). *After the finite Riesz modes are transferred to Z , define*

$$\widehat{K}_{\text{res}}(z) = zI - A_{\text{res}} - C_{\text{res}}(z)^{-1}.$$

If this transform is analytic and polynomially bounded in $\text{Re } z > -\gamma$, then the residual kernel decays like $e^{-\gamma' t}$ for every $\gamma' < \gamma$. Adding the finite Z modes reconstructs the complete resolved correlation.

Proof. The sign follows by transforming the Volterra equation and solving for $\widehat{K}$. Move the Bromwich contour from a positive line to $\text{Re } z = -\gamma'$. Polynomial bounds and one integration by parts control the horizontal pieces. The residues crossed by the contour are precisely the modes already represented in Z . $\square$

Theorem 1.7 (History Doob transform). *The positive transfer eigenfunction on the natural extension defines a normalized Doob kernel on past histories. Its suspension semigroup is intertwined with $\mathcal{P}_t$, and the conditional pressure, enhanced homogenization, and causal-memory construction are compatible under this intertwining.*

Proof. Normalize the transfer operator by the eigenvalue and conjugate by the positive eigenfunction. The resulting one-step kernel is g_{Ψ} and therefore acts on the same past filtration as Theorem 1.2. Conjugacy passes to the suspension renewal formula and to the compressed resolvent. The martingale Poisson equation and the memory Schur complement are invariant under this bounded conjugacy. $\square$

SCOPE AND VERIFICATION STATUS

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